

Global Retail Sales Analysis

A Python-Based Data Analytics Project

Data Source: Kaggle - Global Retail Sales Dataset

1. Executive Summary

This report analyzes 5,000 retail orders across 13 countries and 3 sales channels to understand what is driving profit and where the business can improve. The data covers the period from 2023 to 2025 and includes information on product categories, discount levels and monthly sales trends.

Key findings show that USA, India and Pakistan are the top revenue markets, Retail Store is the most profitable channel per order despite Online having higher total revenue, and the business experiences a significant drop in performance in the second half of the year. The discount analysis reveals that higher discounts are linked to lower revenue and profit per order, and that discounts are not being applied with any clear category or channel-based logic.

2. Business Problem Statement

The company sells products across multiple countries, channels and categories but does not have a clear picture of where its profit is actually coming from. Revenue numbers alone do not tell the full story - a channel or country can show high sales while still underperforming on profitability.

This analysis was built around three core questions:

- Which countries, product categories and sales channels are the most profitable?
- Is the current discount strategy helping the business or reducing what it earns per order?
- Are there seasonal patterns in revenue and profit that the business should be planning for?

The goal is to answer these questions using data and translate the findings into practical recommendations that the business can act on.

3. Data Preparation

- The dataset contained 5,000 rows and 16 columns with no missing values and no duplicate records, so no data removal or imputation was required.
- The Order_Date column was converted from string to datetime format to enable month-wise and year-wise time series analysis.
- A Profit_Margin % column was created by dividing Profit by Revenue and multiplying by 100, to measure how efficiently each order converts revenue into profit.
- A Discount_Tier column was engineered by grouping Discount_Percent into three buckets - Low (0-10%), Medium (10-20%) and High (20-25%) - to make discount-level comparisons easier to read and interpret.
- Month and Year columns were extracted from Order_Date to support monthly trend analysis and year-on-year comparisons across the dataset period.

4. Exploratory Data Analysis

4.1 Category-wise Revenue and Profit Performance

	Product_Category	Revenue	Profit
0	Home & Living	5351720.47	1521121.36
1	Electronics	4984890.43	1453166.66
2	Beauty	4957630.94	1391374.53
3	Fashion	4890667.21	1415769.56
4	Sports	4797073.32	1363267.36

- Home and Living leads in both revenue (5.35M) and profit (1.52M), making it the strongest category overall.
- Sports ranks last in both revenue (4.79M) and profit (1.36M), making it the weakest performing category.
- The gap between the highest and lowest category is around 554K in revenue and 157K in profit - not a massive difference, meaning all categories are fairly competitive with each other.

4.2 Revenue Contribution by Country

	Country	Revenue
0	USA	3964301.07
1	India	3828635.98
2	Pakistan	3776641.39
3	UK	2433211.17
4	Germany	1275580.48
5	Turkey	1269259.56
6	UAE	1260817.39
7	Canada	1248359.77
8	South Africa	1247436.42
9	France	1222422.17
10	Singapore	1216114.97
11	Japan	1120517.08
12	Australia	1118684.92

- USA is the top revenue-generating country with 3.96M, followed closely by India (3.82M) and Pakistan (3.77M).
- These three countries together account for a significantly larger share compared to the rest of the markets.
- UK stands out as the only mid-tier country with noticeably higher revenue (2.43M) compared to all others below it.
- The remaining countries - Germany, Turkey, UAE, Canada, South Africa, France, Singapore, Japan and Australia - are all clustered between 1.11M and 1.27M, meaning the company has a low-penetration but evenly spread presence in these markets.

4.3 Sales Channel Performance Overview

	Sales_Channel	Revenue	Profit
0	Online	8502474.37	2414758.08
1	Retail Store	8434342.48	2436574.99
2	Mobile App	8045165.52	2293366.40

- Online leads in total revenue (8.50M) but ranks second in profit (2.41M).
- Retail Store ranks second in revenue (8.43M) but generates the highest total profit (2.43M), making it the most profitable channel overall.
- Mobile App has the lowest revenue (8.04M) and profit (2.29M) across all three channels.
- Retail Store's ability to generate more profit on slightly less revenue than Online points to better cost efficiency in physical stores.

4.4 Profit Margin by Product Category

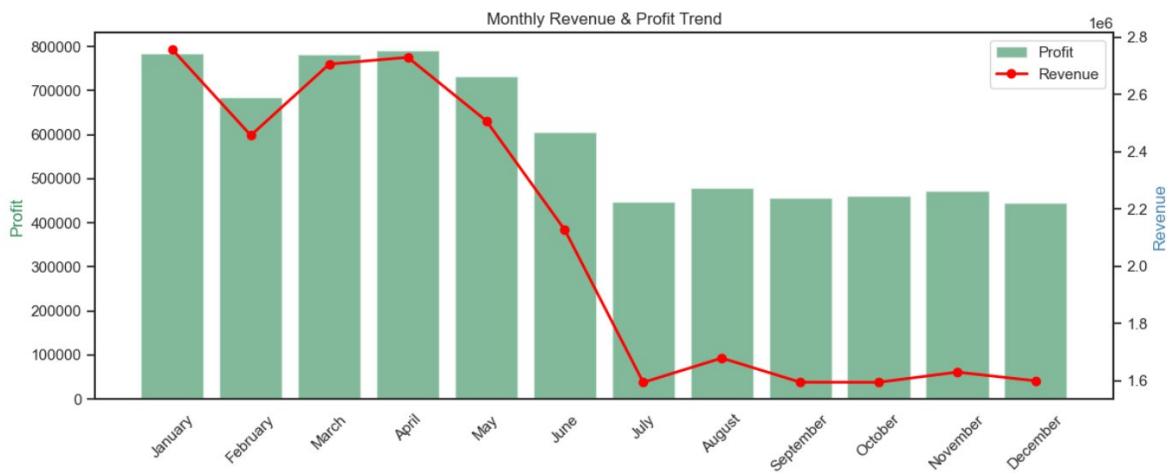
	Product_Category	Profit_Margin %
0	Electronics	28.912
1	Fashion	28.701
2	Sports	28.620
3	Home & Living	28.544
4	Beauty	28.302

- Electronics has the highest profit margin at 28.91%, making it the most margin-efficient category despite not being the revenue leader.

- Beauty has the lowest margin at 28.30%, sitting at the bottom across all five categories.
- The margin range across all categories is only 0.6 percentage points, meaning no category has a serious margin problem or a standout advantage.
- Since all categories operate at a similar efficiency level, margin alone is not a strong enough signal to prioritize one category over another.

4.5 Monthly Revenue and Profit Trend

	Revenue	Profit
Month		
January	2757671.68	785219.22
February	2457590.15	685910.67
March	2706605.10	781502.25
April	2730468.18	791457.66
May	2508746.13	731744.65
June	2127560.13	605748.69
July	1594255.75	448249.60
August	1679339.25	480237.93
September	1595059.72	457191.07
October	1594730.13	460322.22
November	1630209.25	472464.94
December	1599746.90	444650.57



- January is the strongest month with the highest revenue (2.75M) and profit (785K), pointing to strong post-holiday or new year demand.
- April is the second strongest month in profit (791K) despite slightly lower revenue than January, indicating better cost efficiency during that period.
- June marks the start of a sharp decline - revenue drops to 2.12M and profit to 605K, the lowest point in the first half of the year.
- July through December form a consistently weak second half, with revenue staying between 1.59M and 1.67M every month.

- The business is heavily front-loaded - the first six months significantly outperform the last six, which is a seasonal pattern the company needs to plan around.
- Despite the sharp revenue drop in the second half, profit margins hold up relatively well, suggesting the business runs leaner and more cost-efficiently during low-demand months.

4.6 Overall Summary Statistics

=== BASIC SUMMARY ===

Total Orders	: 5000
Total Revenue	: 24,981,982.37
Total Profit	: 7,144,699.47
Avg Profit Margin:	28.62%
Avg Order Value	: 4,996.40
Avg Rating	: 3.02

5. Discount Impact Analysis

5.1 Revenue and Profit by Discount Tier

	Discount_Tier	Revenue	Profit	Profit_Margin %
0	Low	5336.26	1519.24	28.48
1	Medium	4838.32	1393.91	28.82
2	High	4453.29	1282.64	28.61

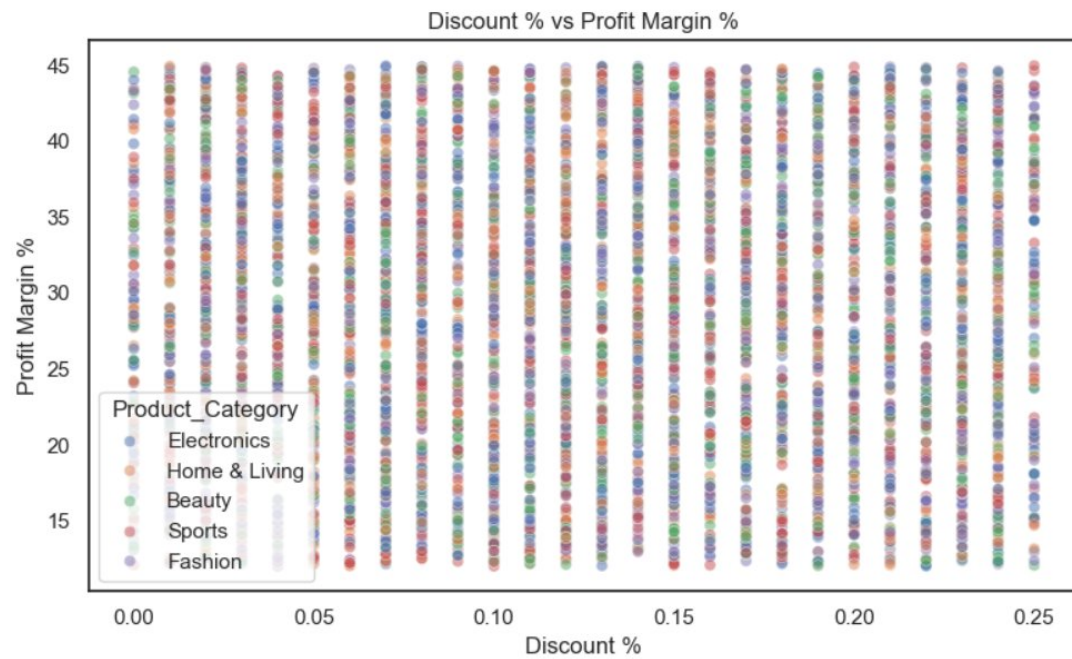
- Both revenue and profit go down as the discount level increases.
- High discount orders generate approximately 16.5% less revenue and 15.6% less profit compared to low discount orders.
- Higher discounts are not bringing in bigger purchases - they are simply reducing what the company earns per order.

5.2 Profit Margin Across Discount Tiers and Categories

	Discount_Tier	Low	Medium	High
Product_Category				
	Beauty	28.656	28.233	27.957
	Electronics	28.110	29.361	29.633
	Fashion	28.885	28.948	28.140
	Home & Living	28.249	29.103	28.059
	Sports	28.553	28.410	29.245

- Discount tiers show negligible variation in profit margin across all product categories, with margins ranging only between 27.9% and 29.6%.
- This suggests discounts are being applied uniformly and are not tailored by category or customer behavior, which points to the absence of a strategic discounting framework.

5.3 Discount Percentage vs Profit Margin - Scatter Plot



- The scatter plot shows no clear pattern between discount percentage and profit margin.
- Dots are spread randomly across all discount levels from 0% to 25%, confirming that changing the discount level has no consistent effect on profit margin.
- This further supports the finding that discounts in this dataset are applied without any strategic logic.

6. Country, Category and Channel Dynamics

6.1 Profitability Comparison Across Countries

	Revenue	Profit	Profit_Margin %
Country			
Germany	5021.97	1472.19	29.75
Australia	4622.66	1347.90	29.53
Japan	5260.64	1534.45	28.96
UK	4599.64	1331.00	28.92
Canada	5095.35	1486.82	28.80
Singapore	5553.04	1537.62	28.68
USA	5141.77	1475.71	28.57
India	5244.71	1492.70	28.55
South Africa	4816.36	1400.03	28.45
UAE	4886.89	1404.37	28.43
France	4701.62	1319.37	28.14
Pakistan	4923.91	1377.16	28.13
Turkey	5016.84	1426.59	28.13

- Germany leads with the highest profit margin at 29.75%, followed by Australia (29.53%) and Japan (28.96%).
- Pakistan, Turkey and France sit at the bottom with margins around 28.13-28.14%.
- The overall spread across all 13 countries is very narrow at just 1.6 percentage points, meaning no country is significantly outperforming or underperforming in margin efficiency.

6.2 Sales Channel Efficiency on a Per-Order Basis

	Revenue	Profit	Profit_Margin %
Sales_Channel			
Mobile App	4953.92	1412.17	28.59
Online	4957.71	1408.02	28.43
Retail Store	5077.87	1466.93	28.84

- Retail Store generates the highest average revenue (5,077.87) and profit (1,466.93) per order with a margin of 28.84%, making it the most profitable channel per transaction.

- Mobile App and Online are very close to each other across all three metrics.
- Despite being the traditional channel, Retail Store is quietly outperforming both digital channels on a per-order basis.

6.3 Country and Category-wise Profit Breakdown

Product_Category	Beauty	Electronics	Fashion	Home & Living	Sports
Country					
Australia	53702.03	67763.78	52281.29	55699.42	96745.73
Canada	92429.48	79263.08	58217.09	63042.07	71318.26
France	64324.02	87616.70	76802.20	70659.64	43634.90
Germany	80010.04	60121.89	53791.30	89922.46	90089.31
India	201389.19	242697.19	248976.22	217656.85	178950.19
Japan	31875.12	71817.23	84407.67	73203.08	65534.38
Pakistan	208297.66	180477.48	218834.30	221675.71	226997.17
Singapore	81512.24	60869.61	49236.08	89751.87	55367.97
South Africa	78504.28	72264.29	58884.78	93396.01	59558.23
Turkey	78509.09	77530.76	46046.25	82366.94	76473.89
UAE	74431.76	59045.01	70000.84	85993.86	72856.44
UK	134750.23	156779.71	153254.11	132444.80	126871.30
USA	211639.39	236919.93	245037.43	245308.65	198869.59

- USA and India are the two biggest profit contributors overall across all categories.
- USA dominates in Home and Living (245K) and Fashion (245K), while India leads in Fashion (248K) and Electronics (242K).
- Pakistan ranks third in total profit volume, driven strongly by Sports (226K) and Home and Living (221K).
- Japan and Australia are the weakest contributors, likely due to lower order volumes rather than poor margins.

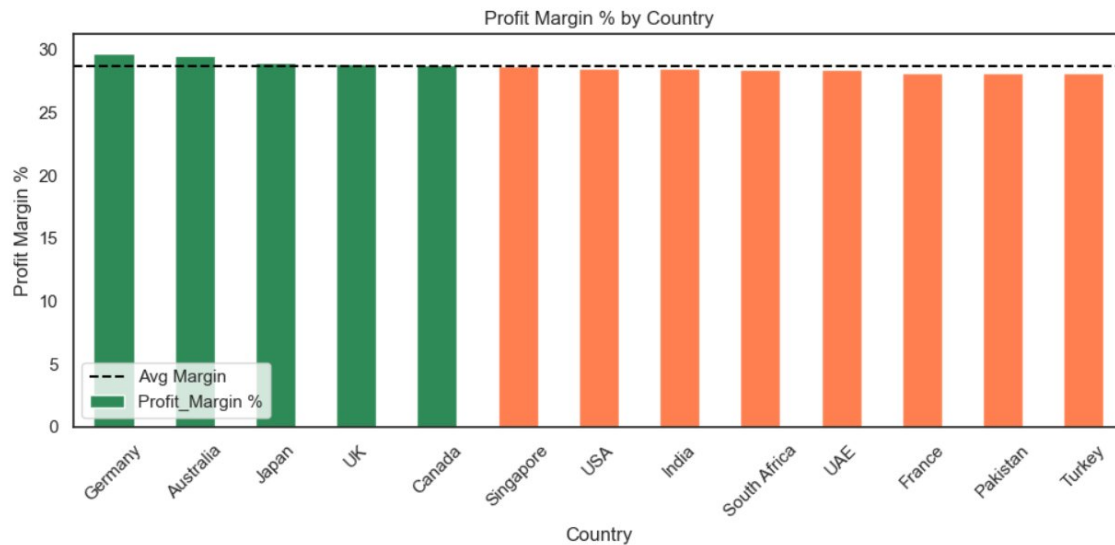
6.4 Best Performing Channel and Category Combinations

Product_Category	Beauty	Electronics	Fashion	Home & Living	Sports
Sales_Channel					
Mobile App	471891.05	468872.94	452765.63	427968.35	471868.43
Online	433007.81	516124.72	471018.88	511565.32	483041.35
Retail Store	486475.67	468169.00	491985.05	581587.69	408357.58

- Retail Store combined with Home and Living is the single highest performing combination at 581K in profit.
- Online with Electronics (516K) and Online with Home and Living (511K) also perform strongly.

- Mobile App generates the most balanced profits across all categories with no single category dominating.
- The weakest combination is Retail Store with Sports (408K), suggesting sports products perform better through online or app-based channels.

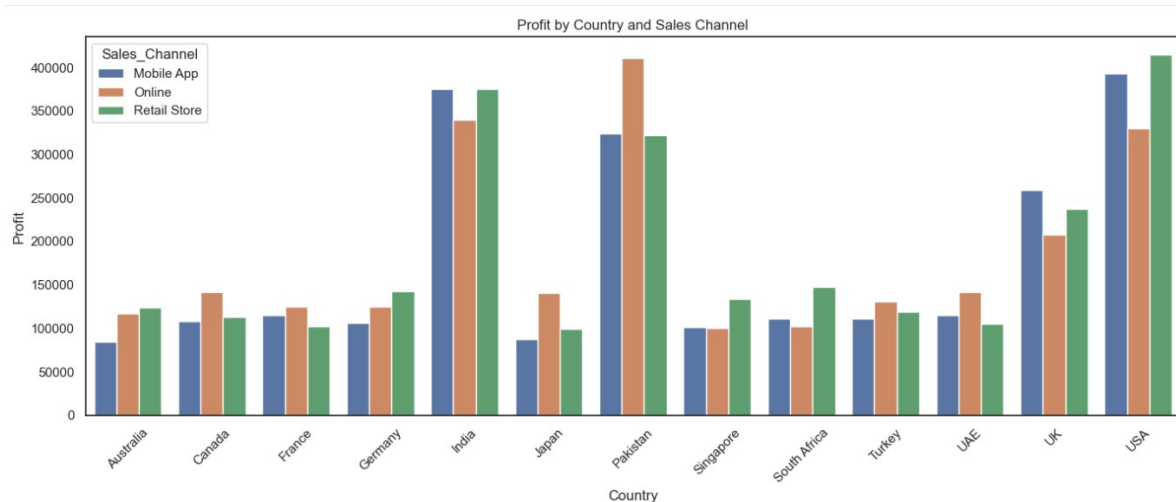
6.5 Profit Margin by Country - Visual Overview



- Only 5 countries - Germany, Australia, Japan, UK and Canada - are above the average margin line.
- The remaining 8 countries fall below the average but the difference between the highest (29.75%) and lowest (28.13%) is barely 1.6%.
- No country is in a critical underperformance zone - the business maintains consistent margins across all global markets.

6.6 Channel Preference Across Countries

Sales_Channel	Mobile App	Online	Retail Store
Country			
Australia	84466.24	117530.85	124195.16
Canada	108823.98	142103.39	113342.61
France	114928.37	125575.32	102533.77
Germany	106586.74	125017.48	142330.78
India	375125.34	339571.23	374973.07
Japan	87288.54	140519.67	99029.27
Pakistan	323734.36	410543.25	322004.71
Singapore	101636.70	100883.58	134217.49
South Africa	111534.93	102835.49	148237.17
Turkey	111134.59	131090.66	118701.68
UAE	115338.83	141659.00	105330.08
UK	259327.60	207456.65	237315.90
USA	393440.18	329971.51	414363.30



- USA and India are the top profit contributors across all three channels with no significant weak spot in either market.
- Pakistan and Japan strongly prefer Online, while Germany and Australia lean towards Retail Store.
- India is the only country where Mobile App and Retail Store are almost equal, showing strong multi-channel adoption.
- Channel preference varies clearly by country, meaning the company needs a region-specific channel strategy rather than a uniform global approach.

7. Key Findings

- Home and Living is the strongest product category by both revenue and profit while Sports is the weakest - though the gap between all categories is small.
- USA, India and Pakistan are the three dominant markets driving the majority of global revenue and profit.
- Retail Store is the most profitable channel both in total profit and profit per order, despite Online having higher total revenue.
- The business is heavily seasonal - the first half of the year consistently outperforms the second half with revenue nearly halving from July onwards.
- Higher discounts are associated with lower revenue and profit per order, yet profit margins stay flat across all discount tiers, indicating discounts are applied without any strategic framework.
- Channel preference varies significantly by country - Pakistan and Japan favor Online, Germany and Australia favor Retail Store and India shows strong multi-channel adoption.
- Retail Store combined with Home and Living is the single most profitable channel-category combination at 581K in profit.

8. Business Recommendations

1. Focus sales investment on USA, India and Pakistan

These three countries drive the majority of total revenue and profit. Marketing budgets, inventory allocation and channel development should be prioritized in these markets before resources are directed towards lower-performing regions.

2. Strengthen the Retail Store and Home and Living combination

Retail Store combined with Home and Living is the most profitable combination in the entire dataset at 581K. The company should ensure strong in-store availability and visibility for Home and Living products, particularly in high-performing markets like USA and Germany.

3. Build a strategic discounting framework

High discount orders generate 16.5% less revenue and 15.6% less profit compared to low discount orders. The company should move away from blanket discounting and implement a category and channel-specific discount policy based on actual purchase behavior data.

4. Address the second half revenue decline

Revenue drops sharply from July onwards every year. The company should plan targeted demand-generation campaigns, seasonal promotions or new product launches specifically for the July-December period to bring more balance to the annual revenue curve.

5. Adopt a region-specific channel strategy

Pakistan and Japan prefer Online, Germany and Australia prefer Retail Store and India performs well across both Mobile App and Retail Store. A single uniform channel strategy will underperform across these varied preferences - the company needs to tailor its channel investment and marketing by region.

6. Invest in Mobile App growth

Mobile App is the weakest channel in both total revenue and profit per order. Given the global shift towards mobile commerce, the company should focus on improving the app experience and driving adoption particularly in high-potential markets like India where mobile readiness is already strong.

9. Conclusion

This analysis looked at 5,000 global retail orders to identify what is driving profitability and where the business has room to improve. The findings show that while margins are healthy and consistent across countries and categories, there are clear gaps in how discounts are applied, how channels are utilized by region and how the business manages its seasonal performance dip in the second half of the year.

Acting on the recommendations in this report - particularly around discount strategy, channel investment by region and second-half demand generation - will help the company protect its margins, allocate resources more efficiently and drive more consistent revenue growth across its global markets.